Arrays

Mësimi 5

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Vargjet

- Vargu është struktura e ruajtjes së të dhënave më e përdorur;
- It's built into most programming languages
- they offer a convenient jumping off place for introducing data structures and for seeing how object-oriented programming and data structures relate to one another
- In this chapter we'll introduce arrays in Java and demonstrate a home-made array class

Bazat e vargut

- There are two kinds of data in Java:
  - Primitive types (such as int and double)
  - Objects.
- In many programming languages (even
- object-oriented ones such as C++), arrays are primitive types,
- In Java they're treated as **objects**.
- Accordingly, you must use the new operator to create an array

Krijo Varg

```
int[] intArray; // përcakton një referencë për
                një varg
intArray = new int[100]; // creates the array,
                and sets intArray to refer to it
```

- Ose mund të përdorni qasjen ekuivalente me një deklaratë të vetme:

```
int[] intArray = new int[100]
```

Krijo Varg

- The [] operator is the sign to the compiler we're naming an array object and not an ordinary variable.
- We can also use an alternative syntax for this operator, placing it after the name instead of the type:

```
int intArray[] = new int[100]; // sintaksë alternative
```

- However, placing the [] after the int makes it clear that the [] is part of the type, not the name.

Qasja në Elementet e Vargut

- Elementet e vargut qasen duke përdorur një numër indeksi në kllapa katrore.
- This is similar to how other languages work:

```
temp = intArray[3]; // merr përmbajtjen e elementit të katërt të vargut
intArray[7] = 66; // fut 66 në qelizën e tetë
```

Shembuj

- Sugjerime ????

High Array

class HighArray

- {
- private long[] a; // referencë për vargun a
- private int nElems; // numri i artikujve të dhënash
- //----------
- public HighArray(int max) // konstruktori
- {
- a = new long[max]; // krijo vargun
- nElems = 0; // asnjë artikull ende
- }
- //----------
- public boolean find(long searchKey)
- { // gjej vlerën e specifikuar
- int j;
- for(j=0; j<nElems; j++) // për çdo element,
- if(a[j] == searchKey) // u gjet artikulli?

High Array

- break; // dil nga cikli para fundit

- if(j == nElems) // shkoi deri në fund?
- return false; // yes, can't find it
- else
- return true; // jo, u gjet
- } // fundi find()
- //----------
- public void insert(long value) // vendos elementin në varg
- {
- a[nElems] = value; // fute atë
- nElems++; // rrit madhësinë
- }

High Array

```
public boolean delete(long value)
{
int j;
for(j=0; j<nElems; j++) // kërko atë
if( value == a[j] )
break;
if(j==nElems) // can't find it
return false;
else // u gjet
{
for(int k=j; k<nElems; k++) // lëviz më të lartët poshtë
a[k] = a[k+1];
nElems--; // zvogëlo madhësinë
return true;
}
} // end delete()
```

High Array

- //----------
- public void display() // shfaq përmbajtjen e vargut
- {
- for(int j=0; j<nElems; j++) // për çdo element,
- System.out.print(a[j] + “ “); // shfaqe atë
- System.out.println(“”);
- }

High Array

- class HighArrayApp
- {
- public static void main(String[] args)
- {
- int maxSize = 100; // madhësia e vargut
- HighArray arr; // referencë për vargun
- arr = new HighArray(maxSize); // krijo vargun
- arr.insert(77); // fut 10 artikuj
- arr.insert(99);
- arr.insert(44);
- arr.insert(55);
- arr.insert(22);
- arr.insert(88);
- arr.insert(11);
- arr.insert(00);
- arr.insert(66);
- arr.insert(33);
- arr.display(); // shfaq artikujt
- int searchKey = 35; // kërko artikullin

High Array

- if(arr.find(searchKey))
- System.out.println("Found " + searchKey);
- else
- System.out.println("Can't find " + searchKey);
- arr.delete(00); // fshi 3 artikuj
- arr.delete(55);
- arr.delete(99);
- arr.display(); // shfaq artikujt përsëri
- } // fundi main()
- } // end class HighArrayApp

High Array

- In the HighArray class, the **find() method** looks through the array for the item whose key value was passed to it as an argument. It returns true or false, depending on whether it finds the item.
- The **insert() method** places a new data item in the next available space in the array. A field called nElems keeps track of the number of array cells that are actually filled with data items. The main() method no longer needs to worry about how many items are in the array.
- The **delete() method** searches for the element whose key value was passed to it as an argument and, when it finds that element, shifts all the elements in higher index cells down one cell, thus writing over the deleted value; it then decrements nElems.

Vargu i Renditur

- Imagine an array in which the data items are arranged in order of ascending key values—that is, with the smallest value at index 0, and each cell holding a value larger than the cell below.
- Such an array is called an *ordered array.*
- Kur fusim një artikull në këtë varg, duhet të gjendet vendndodhja e saktë për
- futjen:
- just above a smaller value and just below a larger one.
- Then all the larger values must be moved up to make room.

Kërkimi Binar me Metodën find()

- The find() method searches for a specified item by repeatedly dividing in half the range of array elements to be considered
- public int find(long searchKey)
- {
- int lowerBound = 0;
- int upperBound = nElems-1;
- int curIn;

Kërkimi Binar me Metodën find()

- while(true)
- {
- curIn = (lowerBound + upperBound) / 2;

- if(a[curIn]==searchKey)
- return curIn; // u gjet

- else if(lowerBound > upperBound)
- return nElems; // can’t find it

Kërkimi Binar me Metodën find()

- else // divide range
- {
- if(a[curIn] < searchKey)
- lowerBound = curIn + 1; // it's in upper half
- else
- upperBound = curIn – 1; // it's in lower half
- } // end else divide range
- } // end while
- } // fundi find()

Kërkimi Binar me Metodën find()

- The method begins by setting the lowerBound and upperBound variables to the first and last occupied cells in the array.
- Setting these variables specifies the range where the item we’re looking for, searchKey, may be found.
- Then, within the while loop, the current index, curIn, is set to the middle of this range.

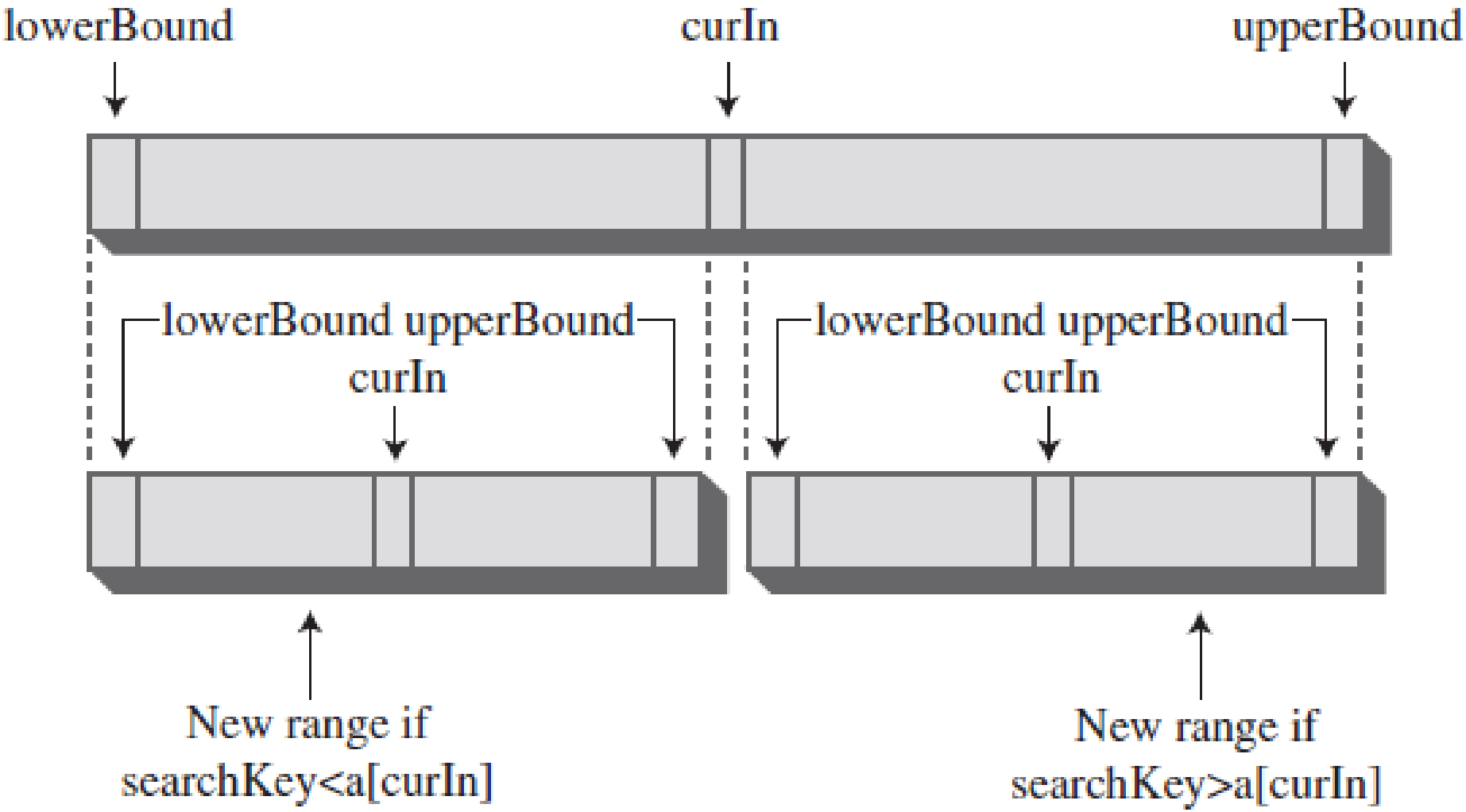
Kërkimi Binar me Metodën find()

- Each time through the loop we divide the range in half. Eventually, the range will get so small that it can't be divided any more.
- We check for this in the next statement:

If lowerBound is greater than upperBound, the range has ceased to exist.

- (When lowerBound equals upperBound, the range is one and we need one more pass through the loop.)

Kërkimi Binar me Metodën find()



Vargjet e Renditura

- Çfarë kemi fituar duke përdorur një varg të renditur?
- The major **advantage** is that search times are much faster than in an unordered array.
- The **disadvantage** is that insertion takes longer because all the data items with a higher key value must be moved up to make room.
- **Deletions** are slow in both ordered and unordered arrays because items must be moved down to fill the hole left by the deleted item.